JOINT UNIVERSITIES PRELIMINARY EXAMINATIONS BOARD

AUGUST 2022 EXAMINATIONS

PHYSICS PRACTICAL **Time Allowed: 2½ Hours**

Answer All Questions.

(i) Plot a graph of d_1 against $(d_1 - d_2)$.

(ii) Determine the slope of the graph.

(iii) If R is the relative density of the solid, show that $d_1 = R(d_1 - d_2)$.

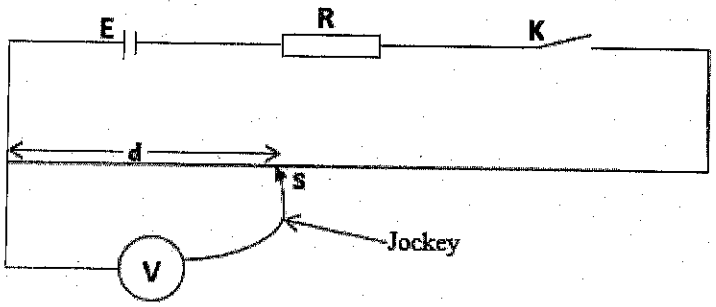
(iv) What does the slope represent?

[Total = 10 Marks]

Question 2

APPARATUS:

Voltmeter, a pair of battery 3V, 2Ω resistor, Key, Potentiometer, Jockey and set of connecting wires



PROCEDURE

I. Connect the circuit as shown in the diagram above.

II. Close the key and use the jockey to make contact with the potentiometer wire such that d = 10 cm

III. Read and record the value of the potential difference V. Evaluate d^{-1} and V^{-1}.

IV. Repeat the experiment for d = 20, 30, 40, 50, and 60 cm respectively. Determine the values of V, V^{-1} and d^{-1} in each case. Tabulate your readings as shown below.

d (cm)	V(v)	$V^{-1}(v^{-1})$	$d^{-1}(cm^{-1})$
10.0			
20.0			
30.0			
40.0			
50.0			
60.0			